HW8

2. $\pi(\xi_1, \xi_2, \xi_3)$, $\pi(x) > 0$, $x = (\xi_1, \xi_2, \xi_3)$, $y = (\eta_1, \eta_2, \eta_3)$

$$p(x, y) = P((\xi_1, \xi_2, \xi_3), (\eta_1, \eta_2, \eta_3))$$

$$= \pi_{1|2,3}(\eta_1 | \xi_2, \xi_3)\, \pi_{2|1,3}(\eta_2 | \eta_1, \xi_3)\, \pi_{3|1,2}(\eta_3 | \eta_1, \eta_2)$$

We write out the conditional distributions with the joined distributions and marginal distributions:

$$\pi_{1|2,3}(\eta_1 | \xi_2, \xi_3) = \frac{\pi(\eta_1, \xi_2, \xi_3)}{\sum_{\xi_1} \pi(\xi_1, \xi_2, \xi_3)}$$

$$\pi_{2|1,3}(\eta_2 | \eta_1, \xi_3) = \frac{\pi(\eta_1, \eta_2, \xi_3)}{\sum_{\xi_2} \pi(\eta_1, \xi_2, \xi_3)}$$

$$\pi_{3|1,2}(\eta_3 | \eta_1, \eta_2) = \frac{\pi(\eta_1, \eta_2, \eta_3)}{\sum_{\xi_3} \pi(\eta_1, \eta_2, \xi_3)}$$

To prove π is the stationary distribution of p, let's prove

$$\sum_{\xi_1} \sum_{\xi_2} \sum_{\xi_3} \pi(\xi_1, \xi_2, \xi_3) \cdot p((\xi_1, \xi_2, \xi_3), (\eta_1, \eta_2, \eta_3)) = \pi(\eta_1, \eta_2, \eta_3)$$

$$\text{Left Side} = \sum_{\xi_2, \xi_3} \left(\sum_{\xi_1} \pi(\xi_1, \xi_2, \xi_3)\, \pi_{1|2,3}(\eta_1 | \xi_2, \xi_3) \right) \cdot \pi_{2|1,3}(\eta_2 | \eta_1, \xi_3) \cdot \pi_{3|1,2}(\eta_3 | \eta_1, \eta_2)$$

$$= \sum_{\xi_2, \xi_3} \pi(\eta_1, \xi_2, \xi_3)\, \pi_{2|1,3}(\eta_2 | \eta_1, \xi_3) \cdot \pi_{3|1,2}(\eta_3 | \eta_1, \eta_2)$$

$$= \sum_{\xi_3} \left(\sum_{\xi_2} \pi(\eta_1, \xi_2, \xi_3)\, \pi_{2|1,3}(\eta_2 | \eta_1, \xi_3) \right) \pi_{3|1,2}(\eta_3 | \eta_1, \eta_2)$$

$$= \sum_{\xi_3} \pi(\eta_1, \eta_2, \xi_3)\, \pi_{3|1,2}(\eta_3 | \eta_1, \eta_2) = \pi(\eta_1, \eta_2, \eta_3) = \text{rhs}$$